RESULTS

Flood Analysis

Model Calibration

The MGLS hydrological flow is one of the most critical inputs for flood analysis. Due to the absence of direct measurements of river discharge during this event, the freshwater inflow to the lagoon system was estimated through a rainfall–runoff approach and subsequently adjusted to reproduce the maximum lagoon water levels observed in the field. The total river discharge flowing into the lagoon system was estimated using a rainfall, runoff model based on the Synthetic Unit Hydrograph (UHT), following DNIT (2005). The data for the UHT were obtained through geoprocessing and from the INMET database: Drainage área = 217 km²; Slope = 0.005%; Concentration time = 6 h; Average Curve Number (CN) = 57. From these parameters, rainfall intensity was calculated for each time interval, and the resulting river discharges were obtained through the Unit Hydrograph Technique (UHT). Figure 1 presents the calculated flow hydrograph.

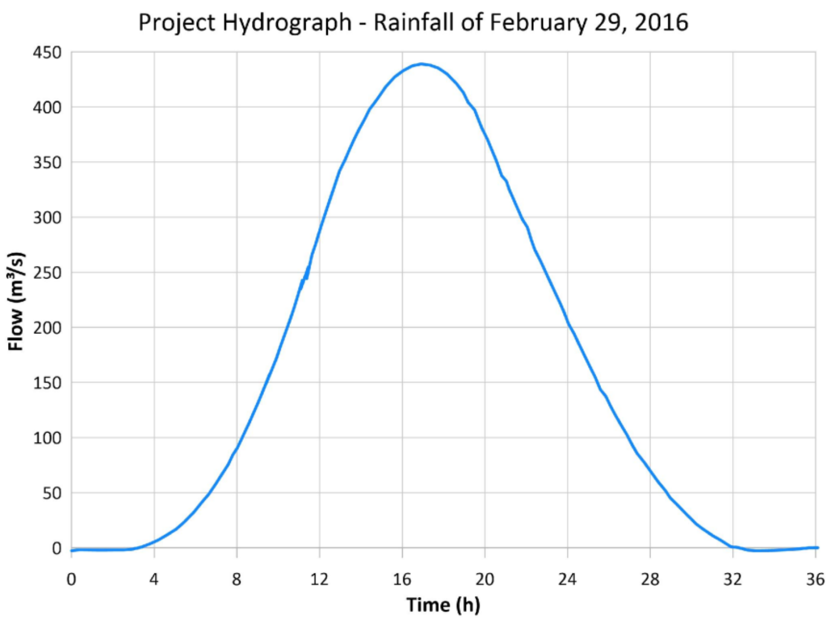

Figure 1: 29 February 2016 calculated rainfall hydrograph.

The resulting river inflow was then used as a starting point for the calibration procedure adopted for the hydrodynamic model, which consisted of slightly adjusting the magnitude of the freshwater inflow hydrograph, so that the simulated lagoon water levels reproduced the maximum water levels observed during the extreme rainfall event of 29 February 2016. Field measurements and observational evidence, including flood marks on buildings and reports from local residents, indicated a 1,0 m maximum water level near the Maricá City Hall and at Barra de Maricá. Since no other measurements are available for that flood event, this indirect measurement is the main data used for model calibration.

The calibration was performed under the Current Situation (CS) with the extreme event scenario, which represents the lagoon system under the existing configuration, without any structural interventions. During this process, only the magnitude of the freshwater inflow hydrograph was adjusted, while the marine boundary conditions remained unchanged.

The Current Situation (CS) scenario, used for water level calibration, showed a hydrodynamic response consistent with the field-observed data. Figure 2 illustrates the hydrograph applied to the model and the evolution of the lagoon water level over 36 hours after the beginning of precipitation, highlighting the monitoring points located near the Maricá City Hall and at Barra de Maricá. One may observe that the lagoon water level rose almost simultaneously in both areas, reflecting a homogeneous system response. As expected, the water level peak coincided with the hydrograph duration, indicating that the system's response followed the flood progression. After about 18 hours of the start of the rainfall, the emergency water level (0.6 m) was reached in both locations. The maximum modeled water level occurred around the 30th hour, with values of 1.01 m near the Barra Channel and 1.02 m near the City Hall, as Figure 2 shows. These results are consistent with field information, including resident reports, journalistic records, and flood marks, thereby confirming the reliability of the adopted model.

It should be noted that this inflow calibration was specifically intended to reproduce observed flood water levels during an extreme rainfall-dominated event. Subsequent analyses of residence time and water renewal focus on comparative scenario performance, and not on absolute discharge values, thereby reducing sensitivity to uncertainties associated with the calibrated inflow magnitude.

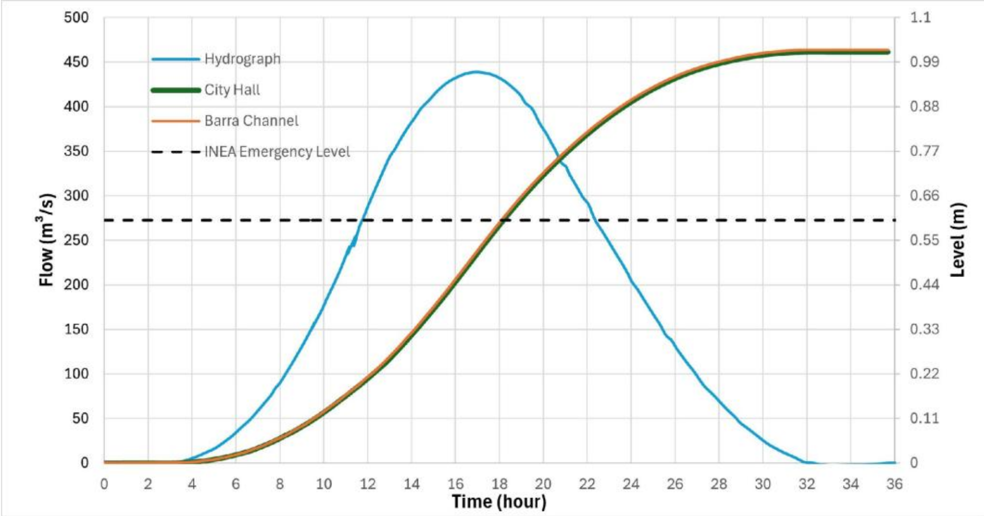

Figure 2: Variation in the level of the lagoon system for the analysis point of the city hall and the coastal channel, together with the calibrated total river flow hydrograph.

Variation of the lagoon level during extreme events

Figure 3 and Figure 4 present the water level results for all simulated scenarios at Barra de Maricá and at the Maricá City Hall, respectively. In the CS scenario, one notices a rapid and significant rise in the lagoon water level within the first 24 hours, reaching approximately 1.02 meters in both locations. In both cases, the water levels remained above the 0.60 m emergency (EWL) threshold even after 15 days, decreasing about 15 cm during this period. This fact highlights the inefficiency of the current configuration of the lagoon system, with only one tidal channel, in properly draining stormwater, even after the flood peak has passed.

The proposed structural interventions scenarios show notable variations in the MGLS's response. In PS1 and PS2 the results are similar. At both analysed locations, the lagoon levels reached between 0.95 m at the beginning of the simulation and then gradually decreased, crossing the emergency threshold around the third day. Despite the notable improvement compared to the current configuration, the water levels remain above 0.60 m for two days. Thus, the Costa Channel by itself considerably improved the the drainage of the extreme rainfall river discharges. However, it was not capable of completely preventing the flooding event.

The PS3 alternative shows a more satisfactory performance. The maximum water level at Barra de Maricá do not exceed 0.35 m and now exhibit regular water level oscillations, associated with the tidal cycle, indicating effective communication with the ocean. In the City Hall region, although the initial values are slightly higher, there is a sharp reduction, with levels remaining within a safe range between 0.30 and 0.45 m. This scenario demonstrates that the stabilization of the Barra Channel, even when implemented alone, provides a substantial improvement in the MGLS's drainage capacity, maintaining levels well below the EWL.

The PS4 scenario, which combines all previous interventions, presented the best performance in flood mitigation among all analyzed scenarios, with results slightly better than PS3. This scenario shows maximum lagoon levels below 0.30 m at both Barra de Maricá and City Hall.

Finally, the Current Dry Situation (CDS) is presented as a reference, highlighting the usual behavior of the MGLS under regular circumstances. It can be observed that in both the Canal da Barra and City Hall regions, lagoon levels remain practically constant over time, with no significant variations associated with the tidal cycle. This behavior confirms the hydrodynamic stagnation condition of these areas.

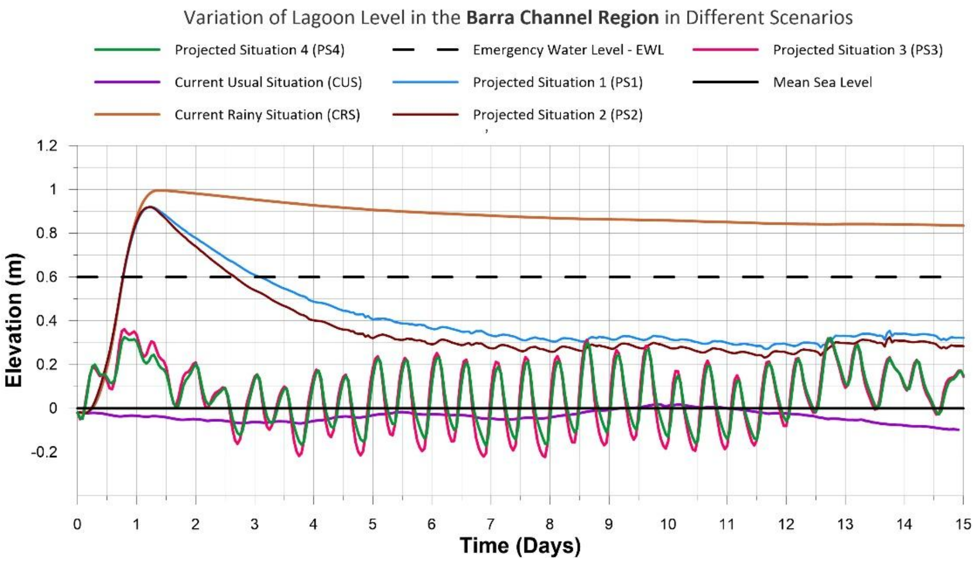

Figure 3: Variation of the lagoon level in the different simulation scenarios at the Barra Channel analysis point.

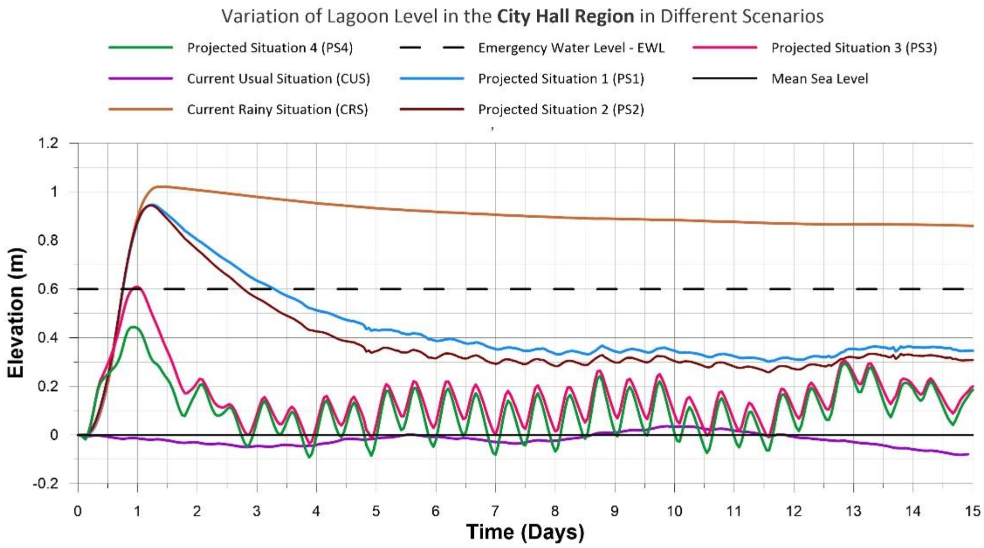

Figure 4: Variation in lagoon level in different simulation scenarios at the City Hall analysis point.

Thus, the results indicate that localized interventions, such as those represented in PS1 and PS2, have a relevant effect, as they reduce flood levels and allow faster drainage of floodwater; however, they are not sufficient to prevent flooding in the city. On the other hand, these interventions would be justified to improve the environmental quality of the Costa Channel and the efficiency of the urban drainage systems in the Itaipuaçu region. The stabilization of the Barra Channel acts as the most effective solution for flood mitigation.

Barra Channel Water Depth Analysis

To determine the most appropriate water depth for the Barra Channel, flow analyses were performed under extreme rainfall conditions using different water depths, considering the PS4 scenario.

As Figure 5 shows, for all the analysed water depths, in the first day the lagoon water level rose between 0.5 and 0.6 m. After approximately three days, the dynamics become dominated by tidal oscillations. Under these conditions, deeper channels exhibit greater tidal amplitude within the lagoon, reflecting a higher capacity for water exchange with the sea. In particular, the 2.0 m depth allows oscillations of up to 0.6 m, even below mean sea level at certain times, while the 0.4 m depth restricts this variation, resulting in smaller amplitudes. Despite these response differences, the mean water level observed for all scenarios ranges between 0.2 m and 0.3 m above the original lagoon mean water level, which is significantly lower than the

EWL. This demonstrates that all tested water depths would efficiently reduce flood risk in the region.

Therefore, in regard of flood control, it is concluded that a 0.8 m water depth would be the most appropriate alternative for the Barra Channel, since it balances hydraulic efficiency in flood control, environmental benefits from increased circulation, and minimum dredging volume in comparison to the higher depth scenarios.

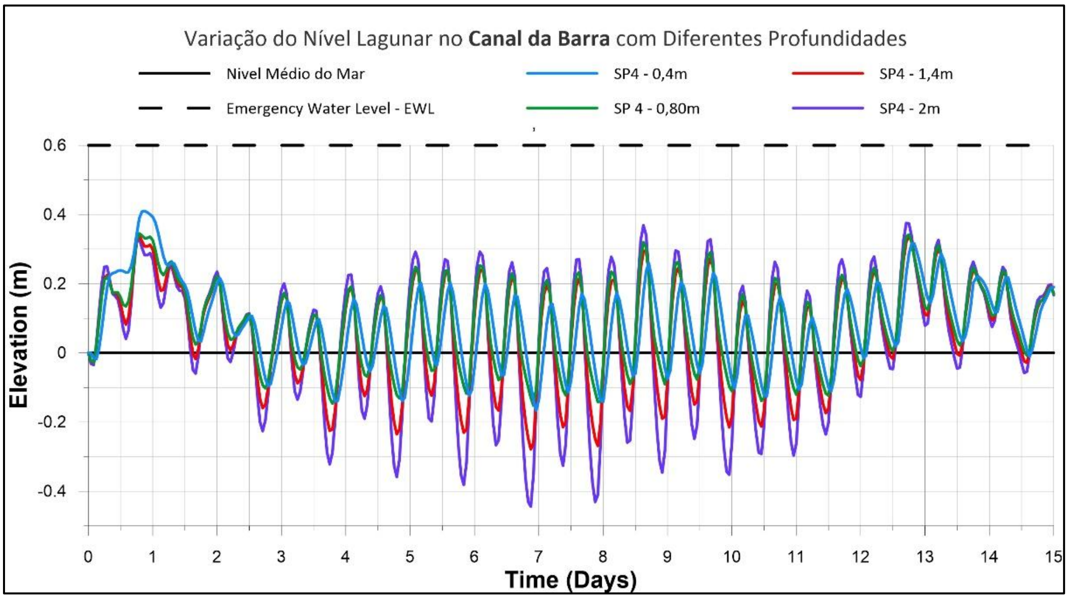

Figure 5: Effect of different water depths of the Barra Channel on the MGLS water level.

Water Renewal and Water Age Analysis

The characteristic hydraulic times, i.e. the Water Renewal Rates and the Water Age, are parameters which indicate the efficiency of mass exchange between the lagoon and the ocean. It is important to notice, however, that higher water renewal rates are not always associated with improved water quality, since water from fluvial inputs may not always be of good quality.

Simulations for the Current Situation revealed high stagnation rates, especially in the Barra Lagoon and Costa Channel, where water ages exceeded 50 days and water renewal rates were below 10%, after 60 days. The Maricá Lagoon showed intermediate performance, with higher renewal near the mouth of the Mumbuca River, while the only portion of the system with efficient renewal was the Guarapina Lagoon, benefitting from its permanent connection to the sea via the Ponta Negra Channel. This pattern was consistent in both summer and winter simulations (Figure 6).

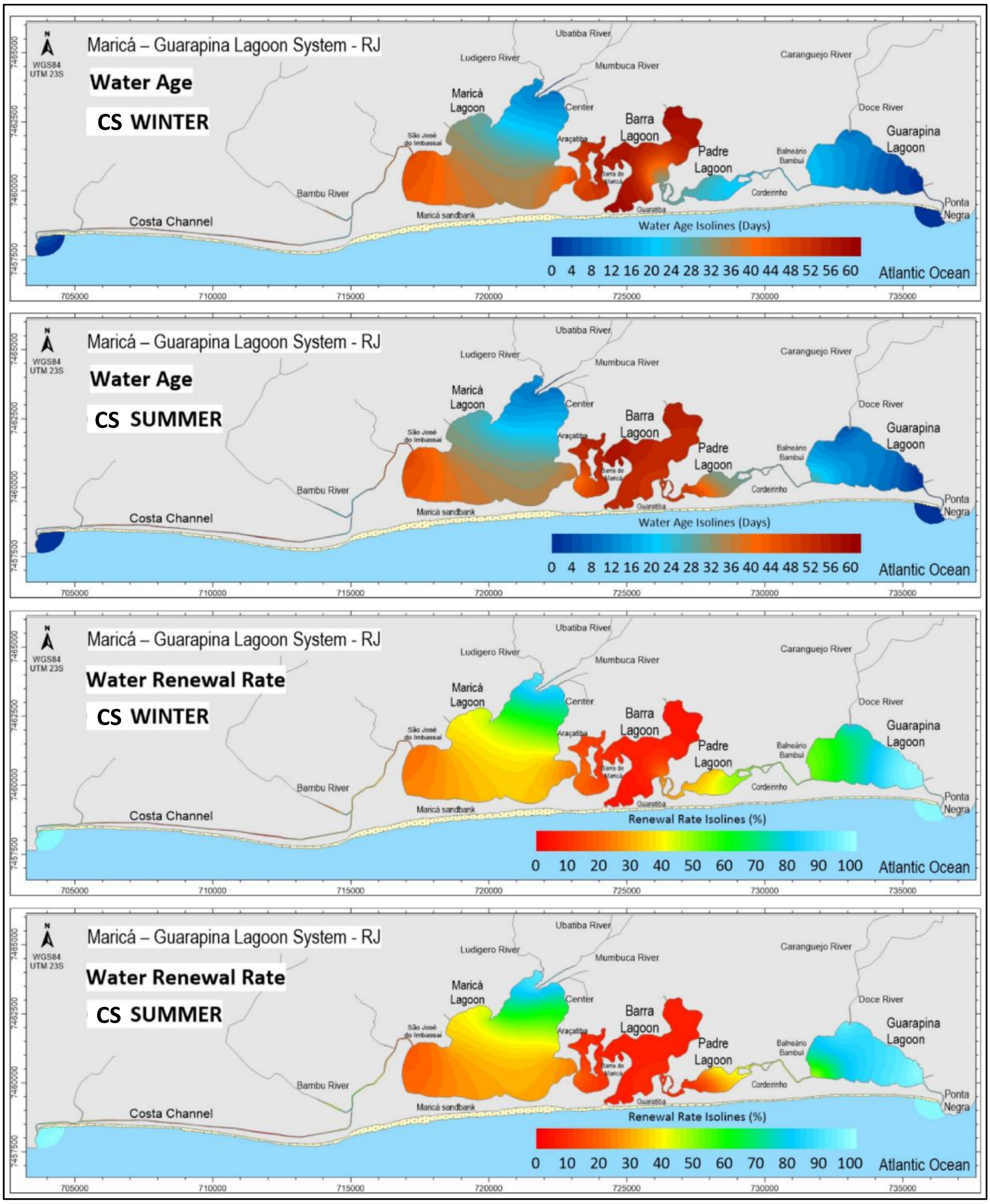

Figure 6: Water Age and Renewal Rate Isolines in the Current Situation, during summer and winter, after 60 days.

In PS1, with clearance and stabilization of the Costa Channel, a significant improvement in the renewal dynamics of the Maricá Lagoon was observed. Most of the lagoon presented water ages around 20 days and water renewal rates of approximately 50% after 60 days, in addition to partial recovery of renewal in the Barra Channel, which reached 100% in some areas. However, this alternative proved effective only for the Maricá Lagoon, without significant impacts on the inner lagoons, such as Barra and Padre. Seasonal differences were minimal, and results remained similar across the two periods analyzed (Figure 7).

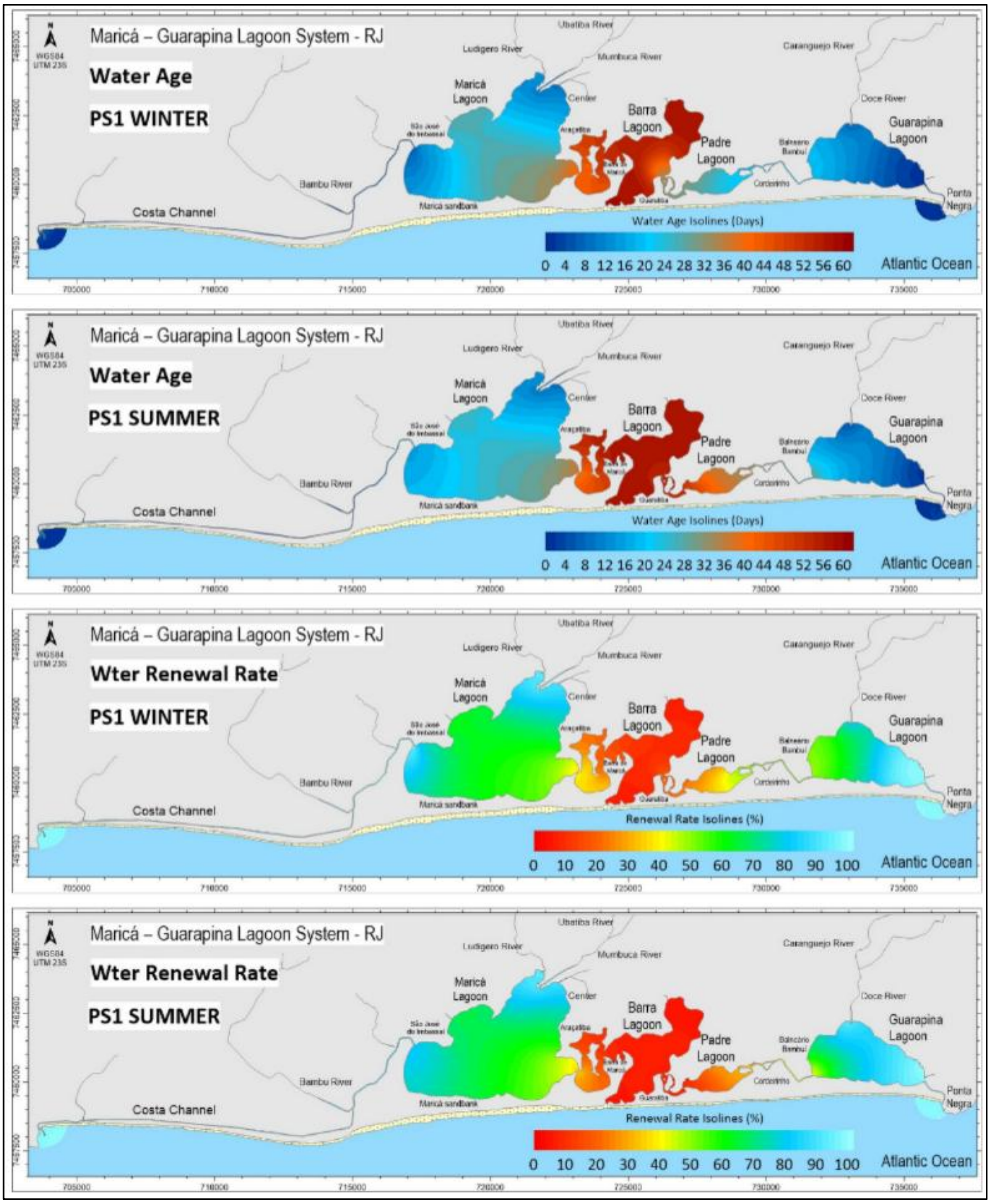

Figure 7: Water Age and Renewal Rate Isolines in Projected Situation 1, during summer and winter, after 60 days.

The PS2 scenario presented superior performance, especially for Barra and Padre Lagoons, areas that, in the current situation, experienced greater difficulty in water renewal. Both lagoons exhibited average water ages of approximately 10 days and water renewal rates above 70% during summer, representing an increase of over 80% compared to the baseline scenario. In winter, due to the influence of storm surges and the resulting differences in water level between the lagoon and the sea, results were even more significant: in the Barra Lagoon, most of the area reached water ages near 5 days and renewal rates exceeding 90%. Figure 8 illustrates these results.

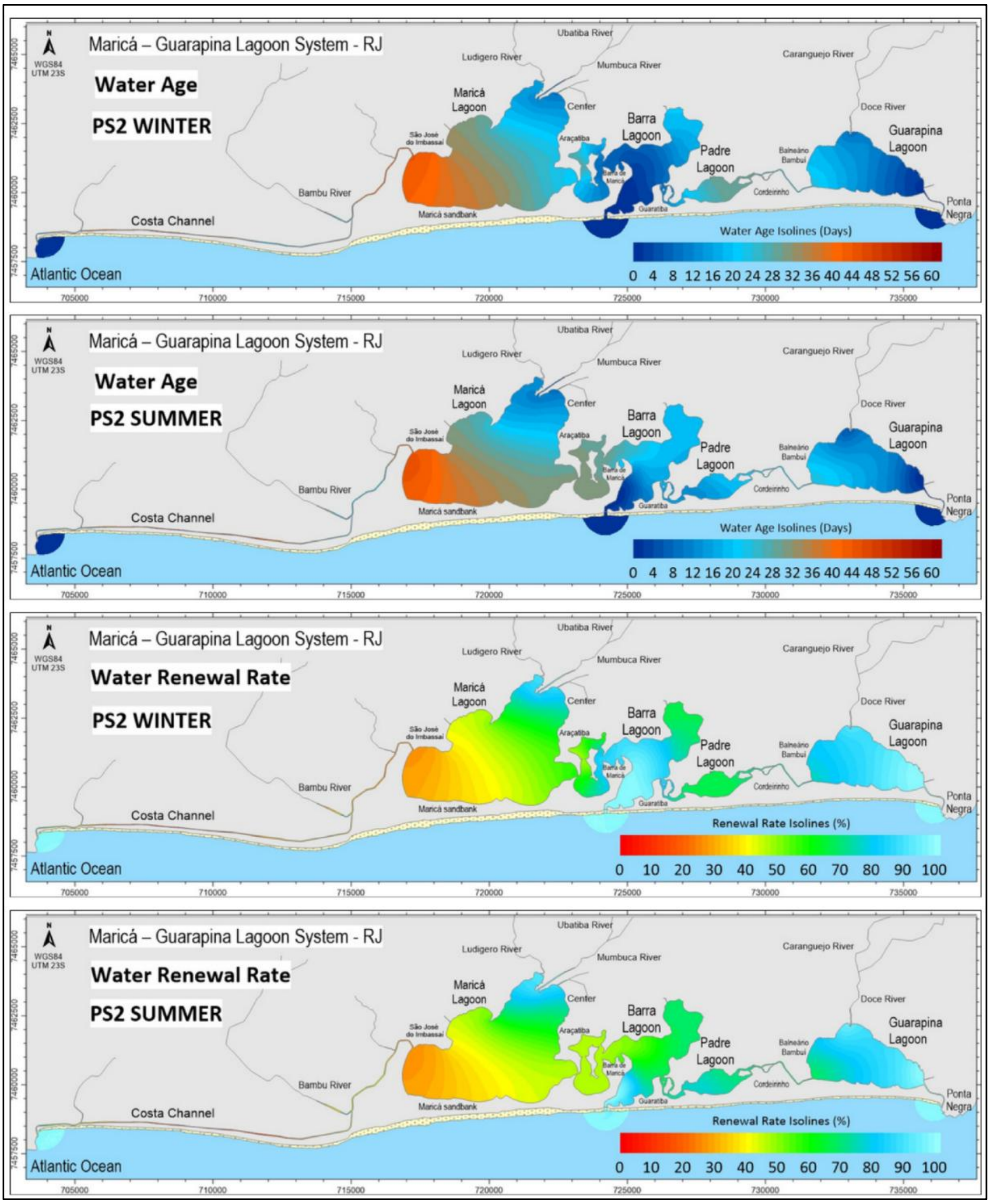

Figure 8: Water Age and Renewal Rate Isolines in Projected Situation 2, during summer and winter, after 60 days.

As Figure 9 depicts, the PS4 scenario delivered a remarkably better result than the other simulated scenarios for both summer and winter. Water renewal rates near 100% after 60 days were observed in practically all regions of the lagoon system, with water ages approaching zero, indicating continuous and efficient exchange between the lagoon and the ocean. This performance reflects an ideal hydrodynamic condition, in which inflow and outflow processes occur frankly, significantly reducing water residence time and promoting environmental recovery of the system.

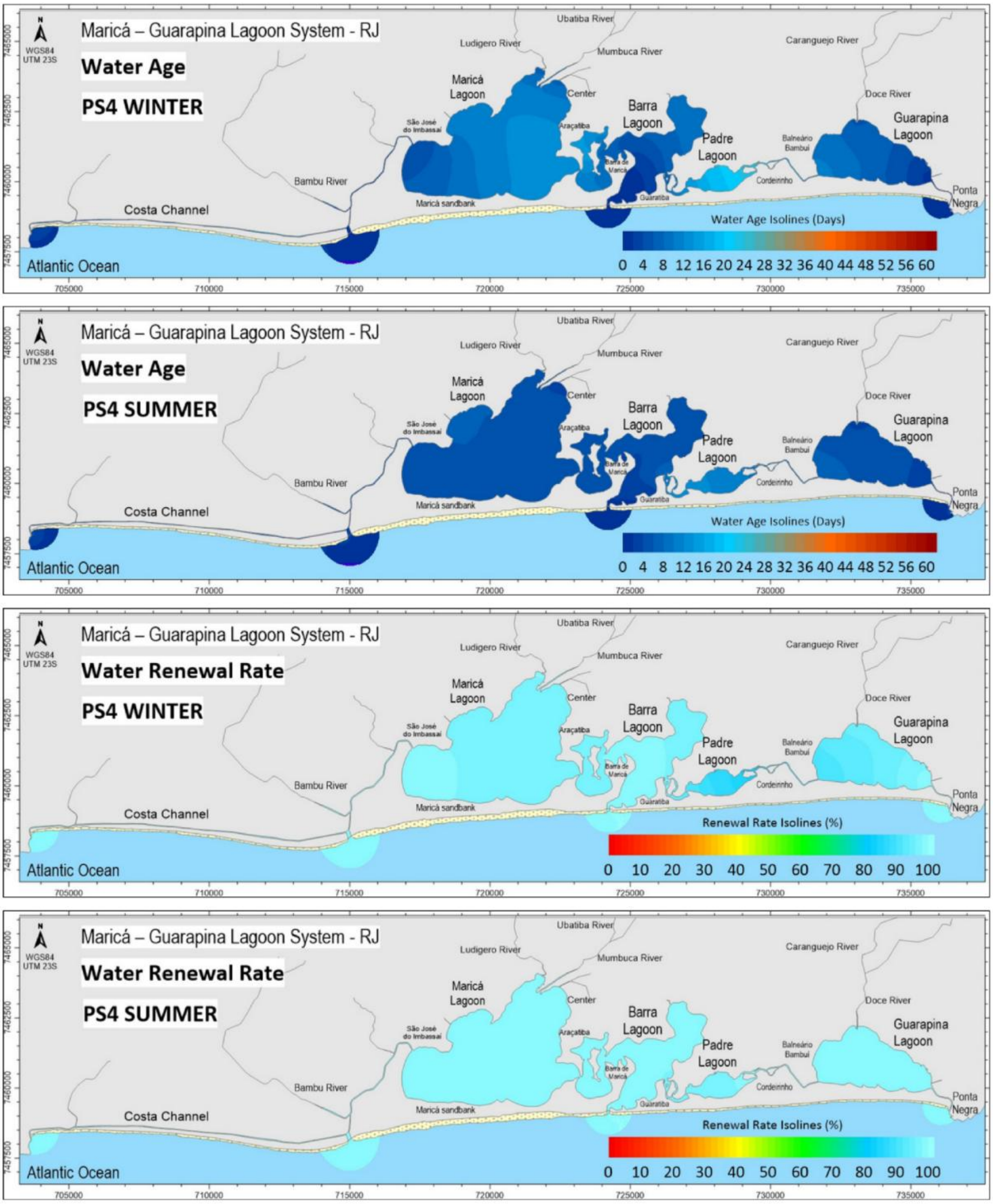

Figure 9: Water Age and Renewal Rate Isolines in Projected Situation 4, during summer and winter, after 60 days.

Finally, to compare the CS with PS4, which is the most effective alternative, Figure 10 presents the temporal evolution of water renewal rates over the 60 simulated days. The results clearly highlight the positive impact of the intervention, revealing progressive improvement in water exchange with the ocean. After 60 days with the channels implemented, the Padre Lagoon reaches approximately 85% renewal, despite being the lagoon that normally exhibits the lowest renewal capacity. By contrast, in the Current Situation (CS) the lagoon that renews the most (Guarapina) reaches only about 70% at the end of the same period. In other words, the channels invert the usual renewal hierarchy: the lagoon that typically renews the least (Padre) surpasses the best-performing lagoon under the CS scenario.

The temporal dynamics further strengthen this conclusion. In the PS4 scenario most lagoons show a rapid increase in water renewal rates during the initial days and then stabilize at high values: the Costa Channel, in particular, goes from virtually 0% renewal in CS to near 100% renewal under PS4. The water renewal rate in the Maricá and Barra Lagoons also rise quickly in PS4 and stabilize at values between 80–90%, while the Guarapina Lagoon in PS4 presents a steadily increase in water renewal rates throughout the simulation, ending substantially higher than in CS. By contrast, under CS several locations remain with low water renewal rates or for much of the simulated period. For example, the Guarapina Lagoon shows modest early renewal and only approaches ~70% late in the simulation.

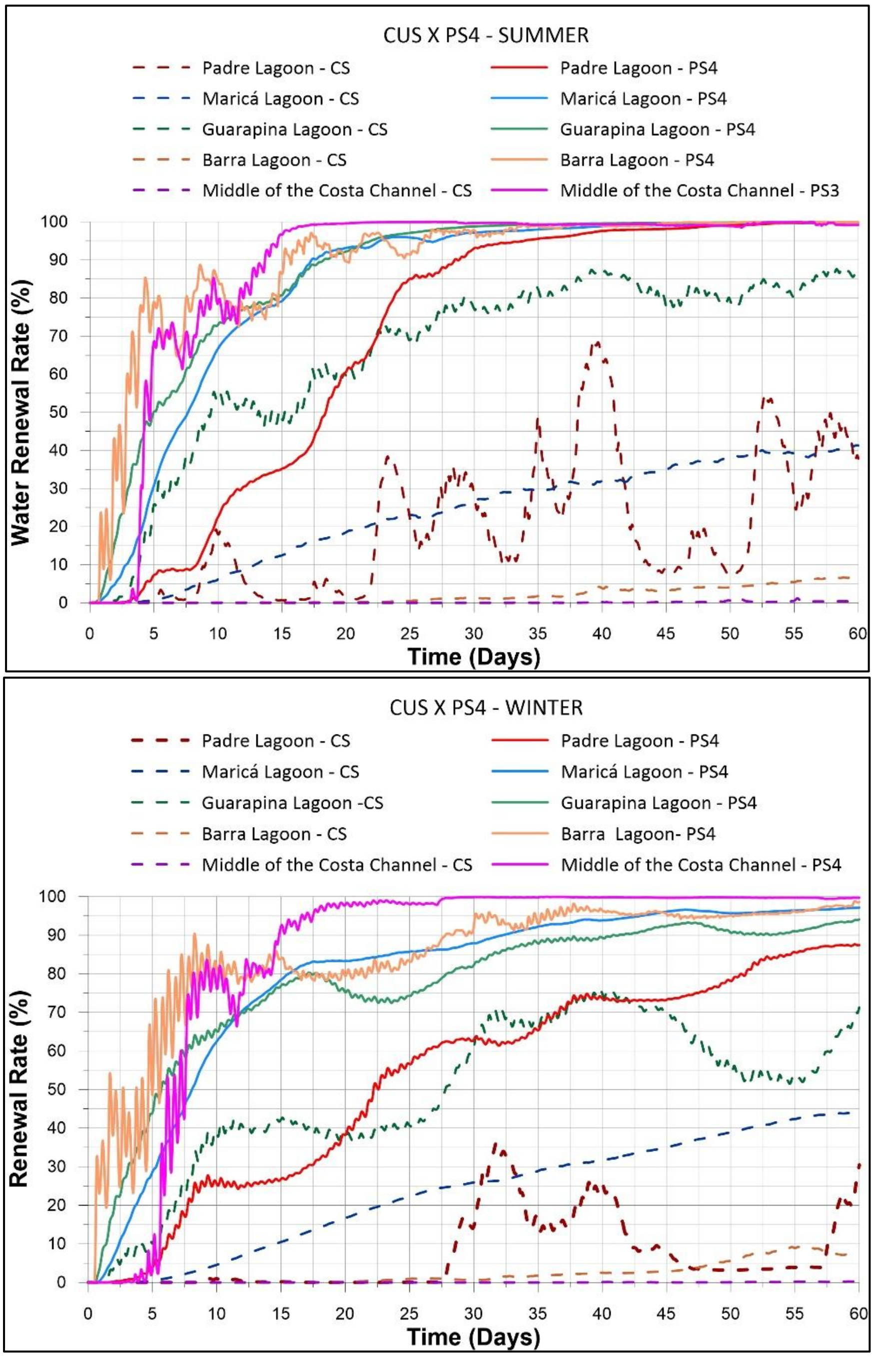

Figure 10: Comparative analysis of the time series of the Current Situation (dashed lines) x Projected Situation 3 (solid lines), including stabilization of the coastal channel, new opening in the middle of the coastal channel and permanent opening of the bar channel.